

Problem 4

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Problem 1 Determine the value of $\lim_{x \rightarrow 0} \left(x^2 \cos \left(\frac{1}{x} \right) \right)$. **EXPLAIN.**

Explanation. $\lim_{x \rightarrow 0} \cos \left(\frac{1}{x} \right)$ does not exist, so we cannot apply the Product law.

We know that for any $x \neq 0$ that $-1 \leq \cos \left(\frac{1}{x} \right) \leq 1$. Multiplying this inequality by x^2 (which is positive) we find that for all x near 0,

$$-x^2 \leq x^2 \cdot \cos \left(\frac{1}{x} \right) \leq x^2$$

The limits of the functions on the left and right sides of this compound inequality are:

$$\begin{aligned} \lim_{x \rightarrow 0} (-x^2) &= 0 \\ \lim_{x \rightarrow 0} x^2 &= 0 \end{aligned}$$

Our explanation:

$\lim_{x \rightarrow 0} \left(x^2 \cos \left(\frac{1}{x} \right) \right) = 0$ by the Squeeze Theorem since $-x^2 \leq x^2 \cdot \cos \left(\frac{1}{x} \right) \leq x^2$ and $\lim_{x \rightarrow 0} (-x^2) = \lim_{x \rightarrow 0} x^2 = 0$.